

Jie Gao

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Malone Postdoctoral Fellow, Johns Hopkins University

Malone Center for Engineering in Healthcare & Center for Language and Speech Processing
NE378, 6225 Smith Ave, The Johns Hopkins at Mt. Washington campus, Baltimore, MD 21209

Research Interests

Human-AI Collaboration; LLM Agents; Multi-Agent Systems; Responsible and Trustworthy AI; AI-Assisted Qualitative Data Analysis; Human Aspects of Software Engineering

My research focuses on **human-AI collaboration for subjective tasks** where ground truth is limited or absent. I design, build, and evaluate collaborative systems along two directions: (1) *Bring AI to Humans*: integrating AI into workflows to help people perform subjective tasks more effectively (e.g., AI-assisted qualitative analysis, codebase comprehension, developer experience); and (2) *Bring Humans to AI*: bringing human oversight to autonomous AI systems to ensure they remain trustworthy and responsible (e.g., AI agent supervision, responsible AI). My work spans Human-Computer Interaction, AI, and Software Engineering.

Employment History

Malone Postdoctoral Fellow

Mar 2025 – Present

Johns Hopkins University | *Malone Postdoctoral Fellowship*

Malone Center for Engineering in Healthcare & Center for Language and Speech Processing

Advisors: Mark Dredze, Ziang Xiao, Chien-Ming Huang

Postdoctoral Associate

Feb 2024 – Feb 2025

Singapore-MIT Alliance for Research and Technology (SMART), MIT

Advisor: Thomas W. Malone

Education

Ph.D. in Information Systems Technology and Design

Sep 2019 – Jan 2024

Singapore University of Technology and Design | *SUTD PhD Fellowship*

Thesis: “Tool Support for Human-AI Collaboration in Qualitative Data Analysis.”

Advisor: Simon Perrault

M.Eng. in Software Engineering (Information Product Design)

Sep 2017 – Jun 2019

Zhejiang University | *Direct admission based on outstanding undergraduate performance*

B.Eng. in Digital Media Technology

Sep 2013 – Jun 2017

Shanghai University | *Top 1% of class*

Key Publications

1. **Jie Gao**, Yue Xue, Xiaofei Xie, Junming Cao, SoeMin Thant, Erika Lee, Bowen Xu. “Understanding Codebase like a Professional! Human-AI Collaboration for Code Comprehension.” *The 34th IEEE/ACM International Conference on Program Comprehension (ICPC 2026)*. *Cross-disciplinary collaboration with Software Engineering*.
2. **Jie Gao**, Yuchen Guo, Gionnieve Lim, Tianqin Zhang, Zheng Zhang, Toby Jia-Jun Li, Simon Perrault. “Col-labCoder: A Lower-barrier, Rigorous Workflow for Inductive Collaborative Qualitative Analysis with Large Language Models.” *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)*.

Highly cited in its domain; Citations: 180+, Jan 2026.

3. **Jie Gao**, Kenny Tsu Wei Choo, Junming Cao, Roy Ka Wei Lee, Simon Perrault. “CoAICoder: Examining the Effectiveness of AI-assisted Human-to-Human Collaboration in Qualitative Analysis.” *ACM Transactions on Computer-Human Interaction (TOCHI 2023)*. Highly cited in its domain; Citations: 100, Jan 2026.
4. Zheng Zhang, **Jie Gao**, Ranjodh Singh Dhaliwal, Toby Jia-Jun Li. “VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping.” *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2023)*. Highly cited in its domain; Citations: 170+, Jan 2026.
5. Nuwan Janaka, **Jie Gao**, Lin Zhu, Shengdong Zhao, Lan Lyu, Peisen Xu, Maximilian Nabokow, Silang Wang, Yanch Ong. “GlassMessaging: Towards Ubiquitous Messaging Using OHMDs.” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT 2023)*.
6. ShunYi Yeo, Gionnieve Lim, **Jie Gao**, Weiyu Zhang, Simon Tangi Perrault. “Help Me Reflect: Leveraging Self-Reflection Interface Nudges to Enhance Deliberativeness on Online Deliberation Platforms.” *Proceedings of the ACM Conference on Human Factors in Computing Systems (CHI 2024)*.
7. Junming Cao, Bihuan Chen, Longjie Hu, **Jie Gao**, Kaifeng Huang, Xuezhi Song, Xin Peng. “Understanding Complexity and Its Impact on Testing in ML-Enabled Systems.” *IEEE International Conference on Software Maintenance and Evolution (ICSME 2023)*.

Peer-Reviewed Short Papers

1. **Jie Gao***, Simret Araya Gebreegziabher*, Kenny Tsu Wei Choo, Toby Jia-Jun Li, Simon Tangi Perrault, Thomas W Malone. “A Taxonomy for Human-LLM Interaction Modes: An Initial Exploration.” *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA 2024)*.
2. **Jie Gao**, Xiayin Ying, Junming Cao, Yifan Yang, Pin Sym Foong, Simon Perrault. “Differences of Challenges of Working from Home (WFH) between China and the US during COVID-19.” *Extended Abstracts of the ACM Conference on Human Factors in Computing Systems (CHI EA 2022)*.
3. Marianne Aubin Le Quéré, Hope Schroeder, Casey Randazzo, **Jie Gao**, Ziv Epstein, Simon Perrault, David Mimno, Louise Barkhuus, Hanlin Li. “LLMs as Research Tools: Applications and Evaluations in HCI Data Work.” *CHI 2024 Workshop Proposal*.
4. Ruyuan Wan, Haonan Wang, Ting-Hao ‘Kenneth’ Huang, **Jie Gao**. “From Noise to Nuance: Enriching Subjective Data Annotation through Qualitative Analysis.” *The 4th HCI+NLP Workshop at EMNLP 2025*. Advising Author.
5. **Jie Gao**, Yuchen Guo, Toby Jia-jun Li, Simon Perrault. “CollabCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis.” *CSCW Demo 2023*.
6. **Jie Gao**, Leijing Zhou, Miaomiao Dong, Fan Zhang. “Expressive Plant: A Multisensory Interactive System for Sensory Training of Children with Autism.” *Ubicomp Poster 2018*.

Honors and Awards

- Malone Postdoctoral Fellowship, Johns Hopkins University 2024
- SUTD PhD Fellowship, Singapore University of Technology and Design 2019
- Excellent Graduate Student, Zhejiang University 2017
- Excellent Undergraduate Student, Shanghai University 2015

Invited Talks & Presentations

- “Human–AI Collaboration for Complex Unstructured Data Analysis” Nov 2025
Western Michigan University
- “From Automation to Human-AI Collaboration: Methods and Applications for Insights Extraction in Unstructured Data” Oct 2025
Malone Center Research Lunch, Johns Hopkins University
- “AI for Text Production and Analysis” — HCIX Research Club Jul 2025
- “Designing Human-AI Teams to Extract Insights from Unstructured Data” Apr 2025
HCI Seminar, Johns Hopkins University
- “Human-Centric Software Engineering in the Era of Large Language Models” Mar 2025
Singapore Management University
- “Extracting Insights from (Un)structured Data: Methods and Applications” Feb 2025
M3S Seminar, Singapore–MIT Alliance for Research and Technology

Academic Service

- **Associate Chair**, ACM Conference on Human Factors in Computing Systems (CHI 2025 & 2026 Main Track, CHI 2022 & 2023 Late-Breaking Work Track),
- **Reviewer**, Empirical Software Engineering (EMSE 2025)
- **Reviewer**, Annual ACM Symposium on User Interface Software and Technology (UIST 2025)
- **Reviewer**, Computers in Human Behavior Reports 2025
- **Reviewer**, International Journal of Human–Computer Interaction (IJHCI 2025, 2026)
- **Reviewer**, ACM Transactions on Interactive Intelligent Systems (TiiS 2024, 2025)
- **Reviewer**, ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW 2022, 2023)
- **Reviewer**, ACM Conference on Human Factors in Computing Systems (CHI 2023, 2024)
- **Reviewer**, International Conference on Artificial Intelligence in Education (AIED 2023)

Visiting Experience

Visiting Ph.D. Student 2023
University of Notre Dame | Advisor: Toby Jia-Jun Li

Visiting Ph.D. Student 2022
National University of Singapore — NUS-HCI Lab | Advisor: Shengdong Zhao

Mentoring Experience

- Yifan Zhang (PhD Student, National University of Singapore) Spring 2026
- Haonan Wang (Master, Johns Hopkins University) Summer–Fall 2025
- Ruyuan Wan (PhD Student, Pennsylvania State University) Summer 2024
- Zhiyao Shu (Master Research Intern, UC Berkeley. Now PhD student at George Mason University) Summer 2024 – Spring 2025
- Xiaoxian Zhang (Master Research Intern, National University of Singapore) Summer 2024
- Erika Lee (Undergraduate Research Intern, UC San Diego) Summer 2024

- Yuchen Guo (Undergraduate Research Intern, SUTD) Fall 2022
- Tianqin Zhang (Undergraduate Research Intern, SUTD) Fall 2022

Teaching Experience

- Teaching Assistant, *50.002: Introduction to Algorithms*, SUTD Fall 2020
- Teaching Assistant, *50.017: Graphics and Visualization*, SUTD Summer 2020